Bayesian Optimization Report: Function 3

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.676	Medium
2	0.676	Medium
3	0.677	Medium
4	0.677	Medium
5	0.678	Medium
6	0.679	Medium
7	0.680	Medium
8	0.680	Medium
9	0.681	Medium
10	0.682	Medium

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.2577	0.2171	0.3519
2	0.2417	0.2106	0.2932
3	0.2299	0.2040	0.2754
4	0.2261	0.1995	0.2677
5	0.2143	0.1912	0.2489
6	0.2043	0.1862	0.2372
7	0.1965	0.1764	0.2305
8	0.1894	0.1720	0.2251
9	0.1823	0.1639	0.2159
10	0.1763	0.1589	0.2155

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.42435843 0.67952155 0.5212976]	-0.042515366483876216	Matern(ν=0.5)	UCB	0.676	None

2	[0.3719064 0.85381306 0.43489009]	-0.030310964488404313	Matern(ν=0.5)	UCB	0.676	None
3	[0.47338521 0.56332675 0.38215493]	-0.026498907468444077	Matern(ν=0.5)	UCB	0.677	None
4	[0.29279666 0.79755813 0.38258937]	-0.02565037251826463	Matern(ν=0.5)	UCB	0.677	None
5	[0.48001325 0.62139479 0.41506393]	-0.024115186703180802	Matern(ν=0.5)	UCB	0.678	None
6	[0.27262911 0.76195581 0.4321501]	-0.024968881258494502	Matern(ν=0.5)	UCB	0.679	None
7	[0.29801777 0.65826725 0.40197347]	-0.024263166006658862	Matern(ν=0.5)	UCB	0.680	None
8	[0.47845681 0.76662918 0.38568842]	-0.022993309351804214	Matern(ν=0.5)	UCB	0.680	None
9	[0.42686367 0.58605643 0.41511757]	-0.024923933560033973	Matern(ν=0.5)	UCB	0.681	None
10	[0.30703704 0.73830743 0.39290292]	-0.022259199990294245	Matern(ν=0.5)	UCB	0.682	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	1.519777
EI	0.024474
PI	0.173969
THOMPSON	1.381241
KG	0.024474
ENTROPY	2.985038
PORTFOLIO	1.491529

Iteration 2

Method	Max Value
UCB	1.179864
EI	0.023110
PI	0.169414
THOMPSON	1.463133
KG	0.023110
ENTROPY	2.674264

PORTFOLIO	1.399770

Iteration 3

Method	Max Value
UCB	1.173215
EI	0.021093
PI	0.164199
THOMPSON	1.450482
KG	0.021093
ENTROPY	2.644065
PORTFOLIO	1.280786

Iteration 4

Method	Max Value
UCB	1.166792
EI	0.019382
PI	0.155262
THOMPSON	1.388638
KG	0.019382
ENTROPY	2.525763
PORTFOLIO	1.246837

Iteration 5

Method	Max Value
UCB	1.160128
EI	0.017258
PI	0.152325
THOMPSON	1.288779
KG	0.017258
ENTROPY	2.372768
PORTFOLIO	1.153513

Iteration 6

Method	Max Value
UCB	1.157160

EI	0.016224
PI	0.148505
THOMPSON	1.271488
KG	0.016224
ENTROPY	2.360943
PORTFOLIO	1.128816

Iteration 7

Method	Max Value
UCB	1.151489
EI	0.014333
PI	0.142954
THOMPSON	1.288664
KG	0.014333
ENTROPY	2.376886
PORTFOLIO	1.065558

Iteration 8

Method	Max Value
UCB	1.151603
EI	0.014171
PI	0.142190
THOMPSON	1.360716
KG	0.014171
ENTROPY	2.344570
PORTFOLIO	1.022949

Iteration 9

Method	Max Value
UCB	1.147050
EI	0.012634
PI	0.136718
THOMPSON	1.298418
KG	0.012634
ENTROPY	2.266243

PORTFOLIO	1.009556

Iteration 10

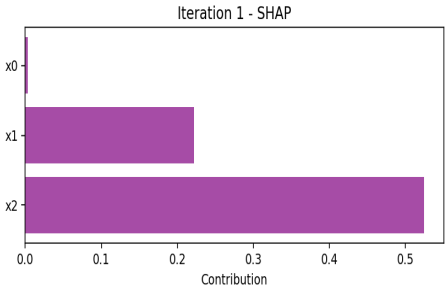
Method	Max Value
UCB	1.145300
EI	0.011935
PI	0.133905
THOMPSON	1.394004
KG	0.011935
ENTROPY	2.343727
PORTFOLIO	0.990172

LOO MSE Summary

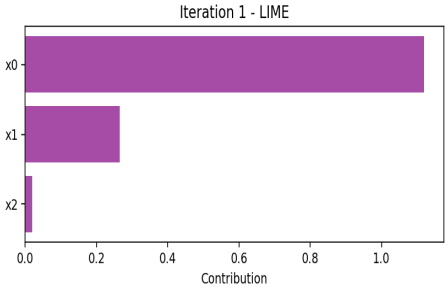
Iteration	LOO MSE
1	0.172127
2	0.160302
3	0.150165
4	0.141966
5	0.134555
6	0.128051
7	0.120418
8	0.107222
9	0.102320
10	0.112969

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



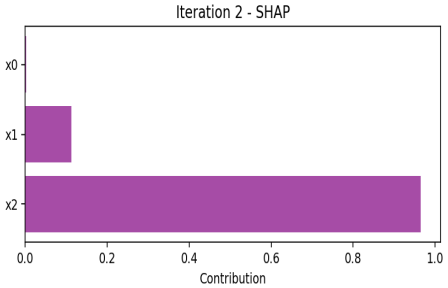
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	0.00319
x1	0.22158
x2	0.52422

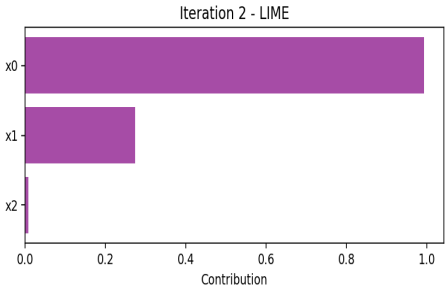
Iteration 1 - LIME Values

Feature	LIME Value
x0	1.11911
x1	0.26483
x2	0.01973

Iteration 2 - SHAP



Iteration 2 - LIME



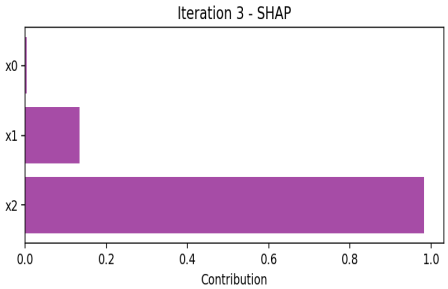
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.00159
x1	0.11186
x2	0.96480

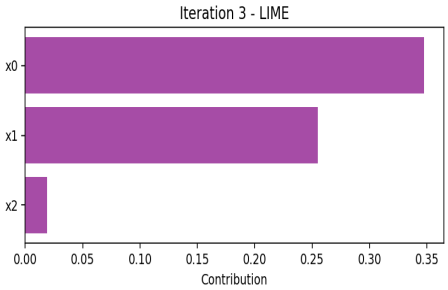
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.99240
x1	0.27284
x2	0.00823

Iteration 3 - SHAP



Iteration 3 - LIME



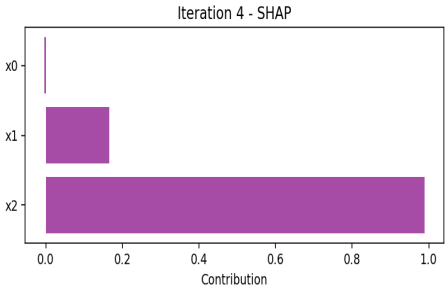
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	0.00407
x1	0.13375
x2	0.98207

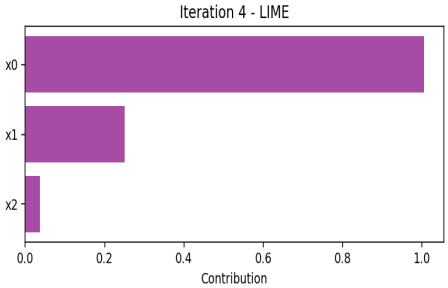
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.34737
x1	0.25489
x2	0.01883

Iteration 4 - SHAP



Iteration 4 - LIME



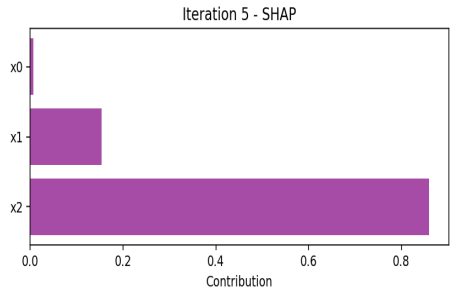
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-0.00368
x1	0.16553
x2	0.98945

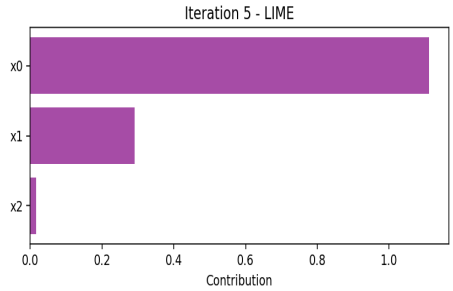
Iteration 4 - LIME Values

Feature	LIME Value
x0	1.00515
x1	0.25027
x2	0.03622

Iteration 5 - SHAP



Iteration 5 - LIME



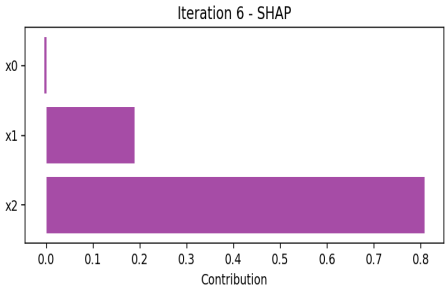
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	0.00677
x1	0.15428
x2	0.85945

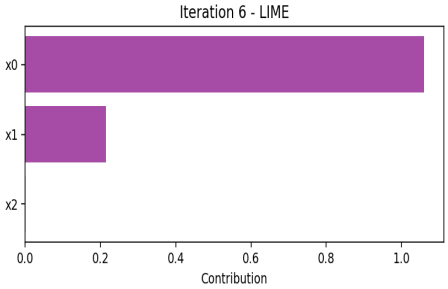
Iteration 5 - LIME Values

Feature	LIME Value
x0	1.11107
x1	0.29098
x2	0.01602

Iteration 6 - SHAP



Iteration 6 - LIME



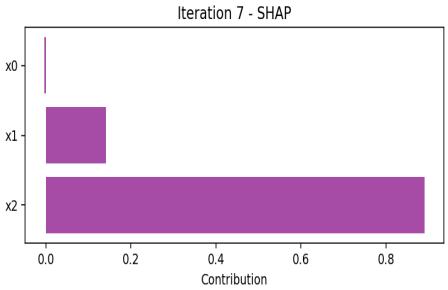
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	-0.00415
x1	0.18901
x2	0.80889

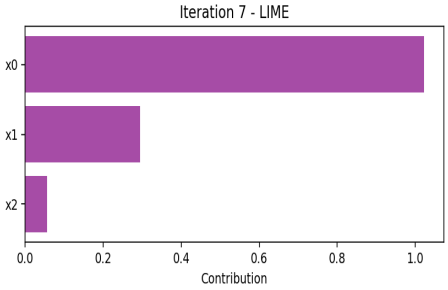
Iteration 6 - LIME Values

Feature	LIME Value
x0	1.05972
x1	0.21431
x2	0.00043

Iteration 7 - SHAP



Iteration 7 - LIME



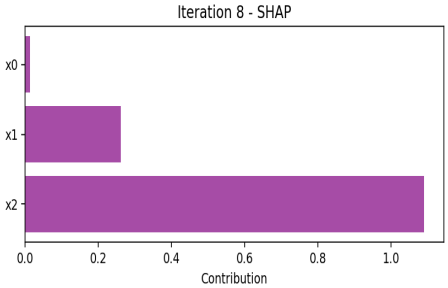
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	-0.00351
x1	0.14168
x2	0.89105

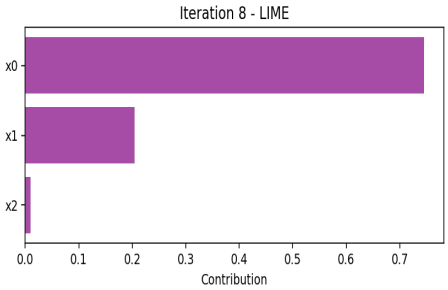
Iteration 7 - LIME Values

Feature	LIME Value
x0	1.02198
x1	0.29392
x2	0.05557

Iteration 8 - SHAP



Iteration 8 - LIME



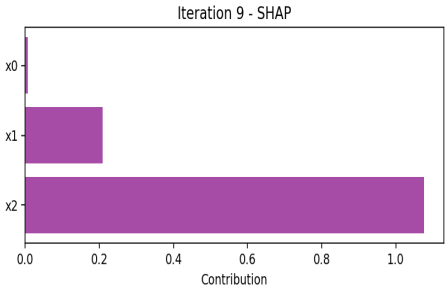
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	0.01222
x1	0.26027
x2	1.08912

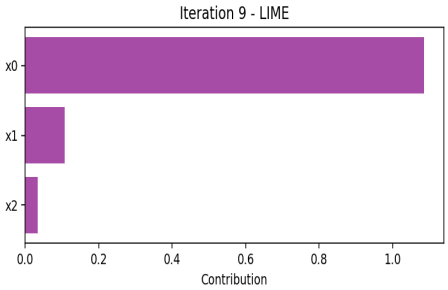
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.74528
x1	0.20404
x2	0.00942

Iteration 9 - SHAP



Iteration 9 - LIME



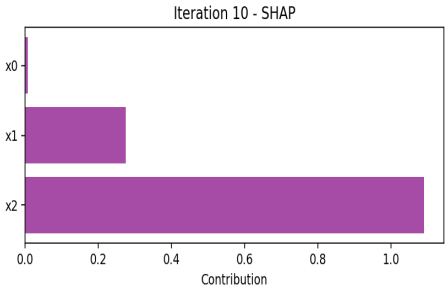
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	0.00651
x1	0.20926
x2	1.07577

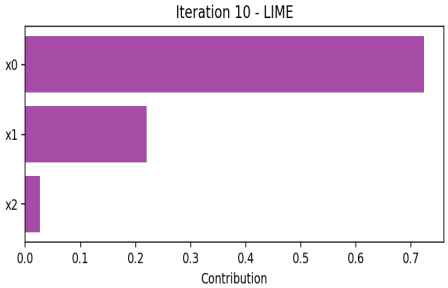
Iteration 9 - LIME Values

Feature	LIME Value
x0	1.08508
x1	0.10706
x2	0.03412

Iteration 10 - SHAP



Iteration 10 - LIME



Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.00733
x1	0.27433
x2	1.09005

Iteration 10 - LIME Values

Feature	LIME Value
x0	0.72331
x1	0.21987
x2	0.02673